



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2019

Bachelor in Computer Applications
Course Title: Data Structures & Algorithms
Code No: CACS 201
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Center:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10 × 1 = 10]

1. Circle (O) the correct answer.

- i) What is the measurement for time complexity of an algorithm?
- a) Counting microseconds b) Counting kilobytes of algorithms
- c) Counting number of key operations d) Counting number of statements
- ii) Which of the following is the result of evaluation of $5 \ 7 \ 4 - * \ 8 \ 4 / + ?$
- a) 5 b) 8 c) 10 d) 17
- iii) What is the recursive formula for post order traversal of binary tree?
- a) Left-Root-Right b) Root-Left-Right
- c) Left-Right-Root d) Right-Left-Root
- iv) What is the number of disk movement in TOH with 4 disks?
- a) 9 b) 14 c) 17 d) 15
- v) What is the Big-Oh of best case complexity of insertion sort?
- a) $O(n)$ b) $O(n \log n)$ c) $O(1)$ d) $O(n^2)$
- vi) How does the rear index incremented in circular queue?
- a) $\text{front} = (\text{rear} + 1) \% \text{SIZE}$ b) $\text{rear} = (\text{rear} + 1) \% \text{SIZE}$

c) $\text{rear} = \text{rear} + 1$

d) $\text{rear} = (\text{rear} - 1) \% \text{SIZE}$

vii) A variation of linked list in which none of the node contains NULL pointer is

a) Singly b) Multiple c) Circular d) Doubly viii) Which of the following data

structure is used in depth first search of graph?

a) Stack

b) Queue

c) Linked List

d) None of the above

ix) Which of the following is true for B-Tree of order M?

a) Leaf nodes should be at different level

b) All the key values within a node must be in descending order

c) Every node has at least M children

d) All non-leaf nodes with M-1 keys must have M number of children

x) Which of the following is not a hash function?

a) Division remainder

b) Folding

c) Chaining

d) Mid square



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Group B

Attempt any SIX questions.

[6 × 5 = 30]

2. What is Data Structure? Show the status of stack converting following infix expression to postfix $P + Q - (R * S / T + U) - V * W$ [1+4]
3. Write binary search. Consider a hash table of size 10; insert the keys 62, 37, 36, 44, 67, 91 and 107 using linear probing. [2+3]
4. What are deterministic and non-deterministic algorithms? Explain greedy algorithm. [3+2]
5. Draw a BST from the string DATASTRUCTURE and traverse the tree in post order and preorder. [3+2]
6. Define circular queue? How does circular queue overcome the limitation of linear queue? Explain. [2+3]
7. What is singly linked list? Write an algorithm to add a node at the beginning and end of singly linked list. [1+4]
8. Define AVL tree. Construct AVL tree from given data set: 4, 6, 12, 9, 5, 2, 13, 8, 3, 7, 11. [2+3]

Group C

Attempt any TWO questions.

[2 × 10 = 20]

9. What is stack? List the applications of stack. Write an algorithm or procedure to perform PUSH and POP operation in stack. [1+2+7]

10. What is heap? Explain quick sort algorithm with Big-oh notation in best case, average case and worst case and trace it to sort the data: 8, 10, 5, 12, 14, 5, 7, 13. [2+2+6]
11. Define graph and tree data structure. Explain breadth first traversal and depth first traversal with example.